STRIDE

Autoencoder-Based Intrinsic Structural Malware Detection

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(HS-SOFT-450)

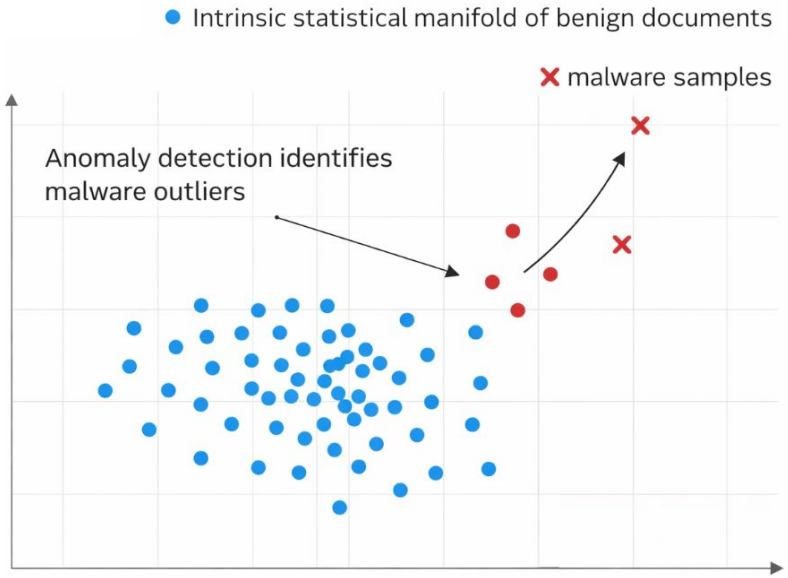

Figure 1: Anomaly Detection

Graphic created by student with Canva, 2026

All figures, graphs, and tables were created by the student researcher

Introduction

Attack Statistics

- 90% of attacks use files [1]
- PDFs are common malware vector
- Malware detection relies heavily on known malware samples
- Zero-day malware bypass signatures

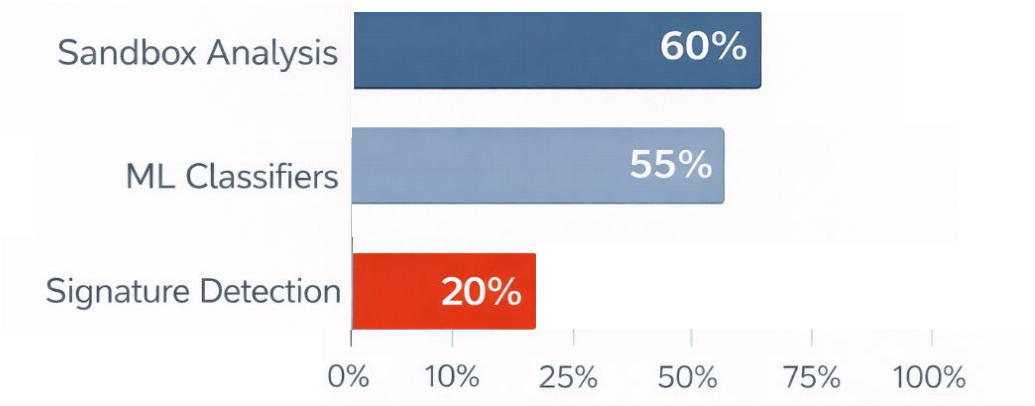

Figure 2: Zero-Day Detection Rates

Graph created by student with Canva, 2026

Sources: Zhang et al. (2022); Raff et al. (2020), MITRE (2023);

[1] Cisco Talos Intelligence Group. (2022).

Methods

Detection Rationale

- Structural PDF feature patterns [1]
- Autoencoder anomaly learning [2]
- Detects zero-day deviations

<u>Goal</u>: To further enhance accuracy of pre-existing malware detectors by using intrinsic metrics rather than comparing to preexisting malware.

[1] Raff et al., 2020

[2] Chalapathy & Chawla, 2019

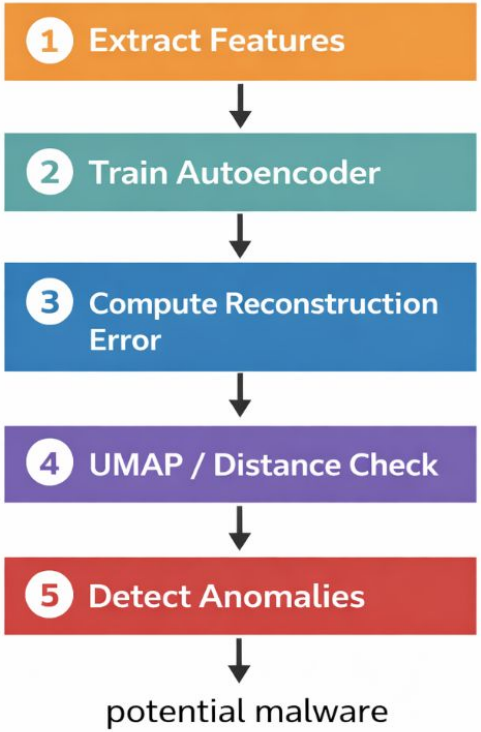

Figure 3: Flow Chart of Procedure

Graphic created by student with Canva, 2026

Benign Dataset Construction

Benign Dataset Overview

- 5000 benign PDFs collected
  - 2500 synthetic generation
  - 2500 web-scraped documents

Synthetic Dataset Design

- Random PDFs →high entropy baseline
- Patterned PDFs → structured repetition
- Low-entropy PDFs → compression extremes

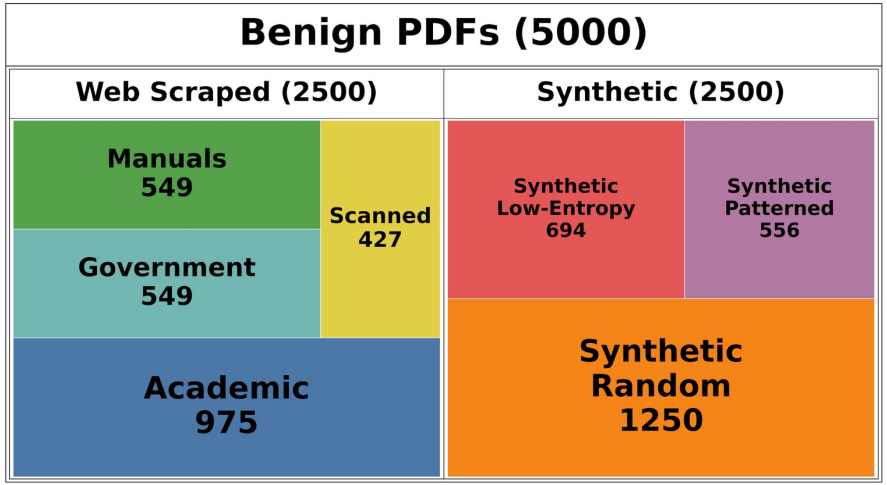

Figure 4: Benign PDF Dataset Diversity Treemap

Graph created by student with matplotlib and squarify, 2026

Structural Feature Extraction

Intrinsic File Structure Metrics

- Shannon entropy [1]
- Lempel-Ziv complexity [2]
- Compression ratio [3]
- File size
- Autoencoder error

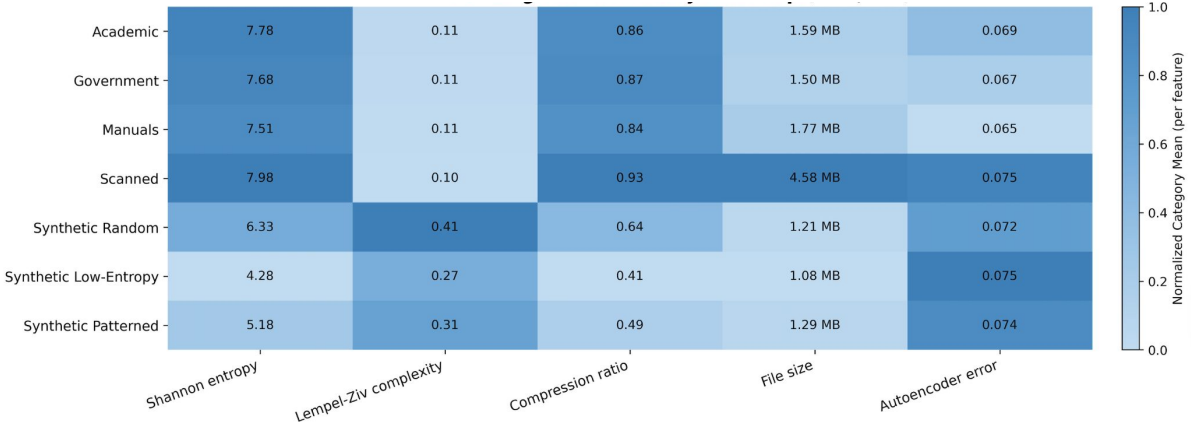

Figure 5: Benign PDF Dataset Feature Extraction Heatmap

Graphic created by student with matplotlib, 2026

[1] Shannon 1948

[2] Lempel & Ziv 1976

[3] Li & Vitányi 2008

Statistical Feature Comparison

- Malware gathered from public datasets like **MalwareBazaar** and **GROUP-IB**
- **Clear difference** in **intrinsic metrics** between benign and malware

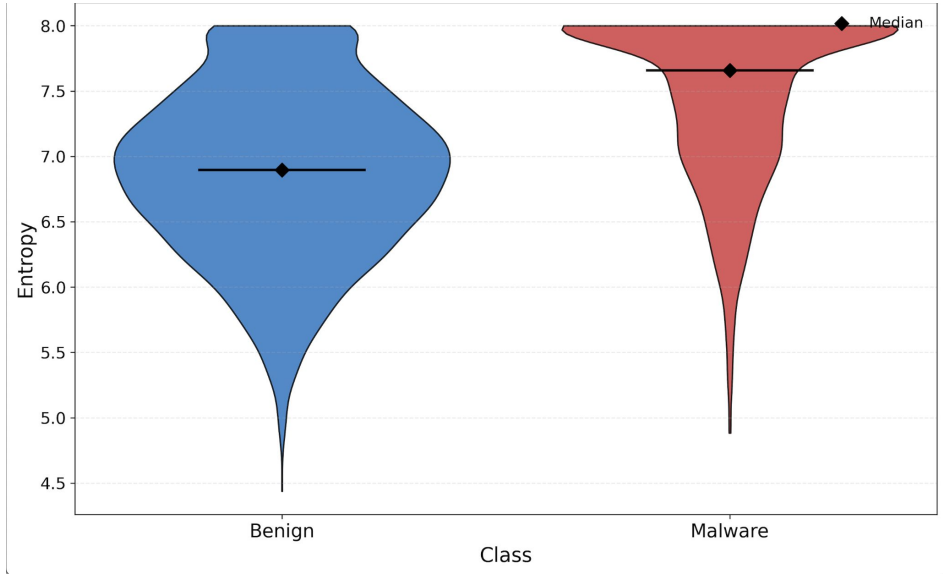

Figure 6: Benign vs Malware Entropy Distribution Violin Graph

Graphic created by student with matplotlib, 2026

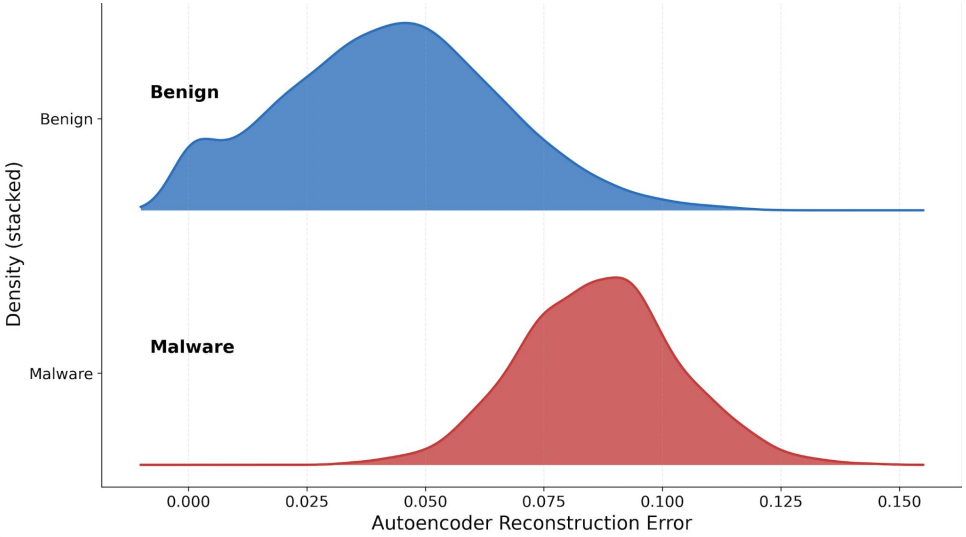

Figure 7: Benign vs Malware Autoencoder Error Ridge Graph

Graphic created by student with matplotlib, 2026

Feature Manifold Visualization

Visualizing A 5D Manifold

- UMAP[1]: dimensionality reduction [2]
- Projects features to 2D
- Reveals benign vs malware structure

[1] Uniform Manifold Approximation and Projection

[2] McInnes, Healy & Melville, 2018

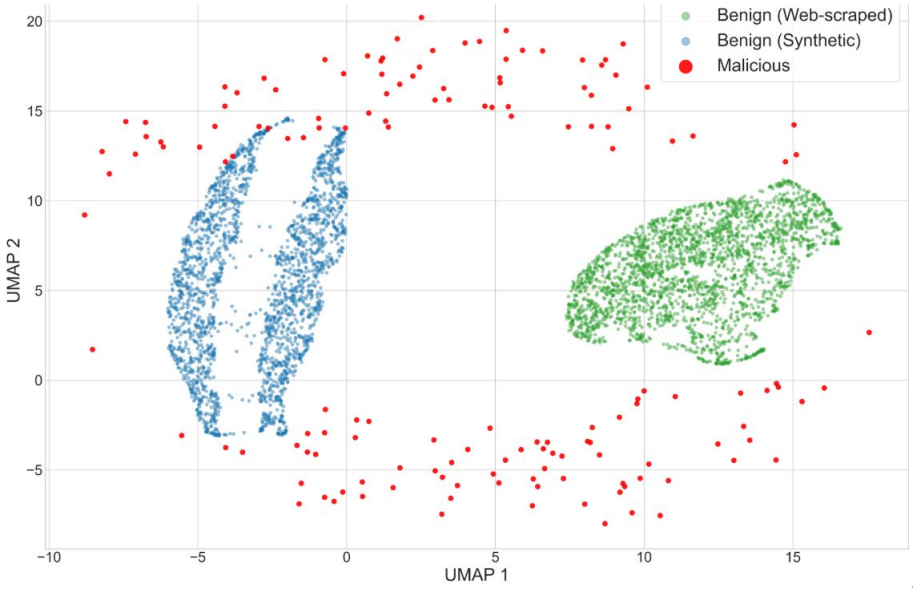

Figure 8: (unsupervised) UMAP shows distinct benign structures with malware scattered across

Graphic created by student with matplotlib, 2026

Nonlinear Malware Classification

Uses a **benign-only Isolation Forest** to create **nonlinear boundaries** and predict class **by region** [1]

[1] Liu, Ting & Zhou, 2008

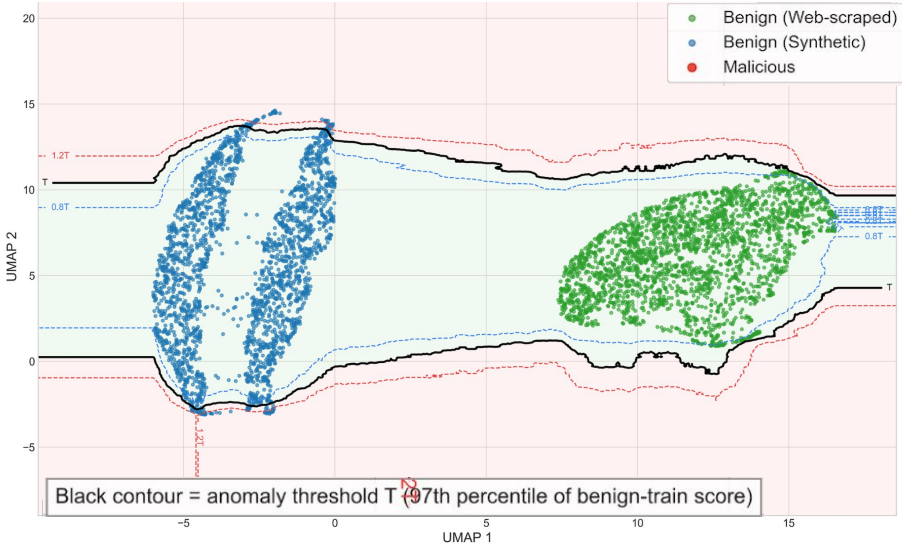

Figure 9: Nonlinear Decision Boundary on UMAP Space

Graphic created by student with matplotlib, 2026

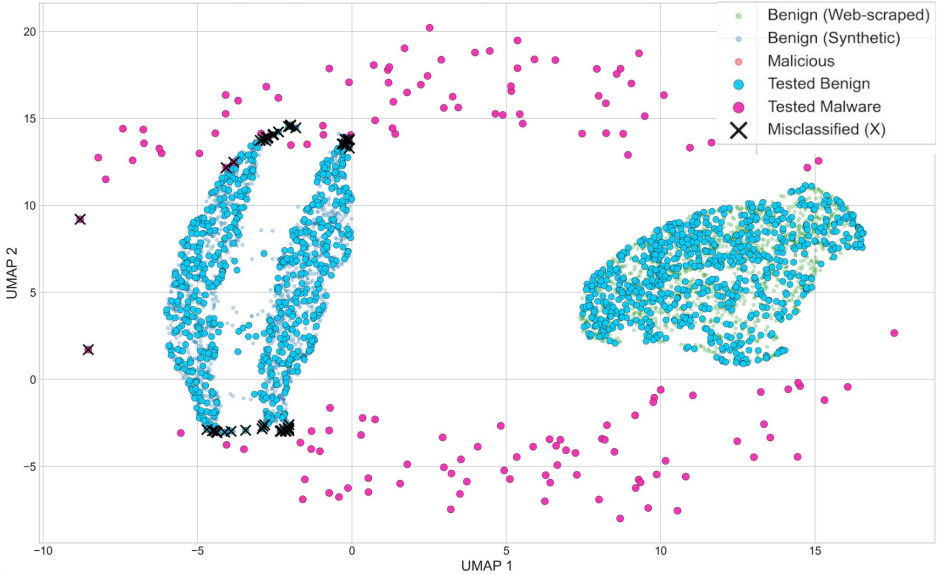

Figure 10: UMAP Test Overlay (**Accuracy = 97.33%**) on malware only, tested benign were part of training as well

Graph created by student with matplotlib, 2026

Results & Analysis

STRIDE has shown to **improve** the metrics of VirusTotal after testing on ~200 malware PDFs

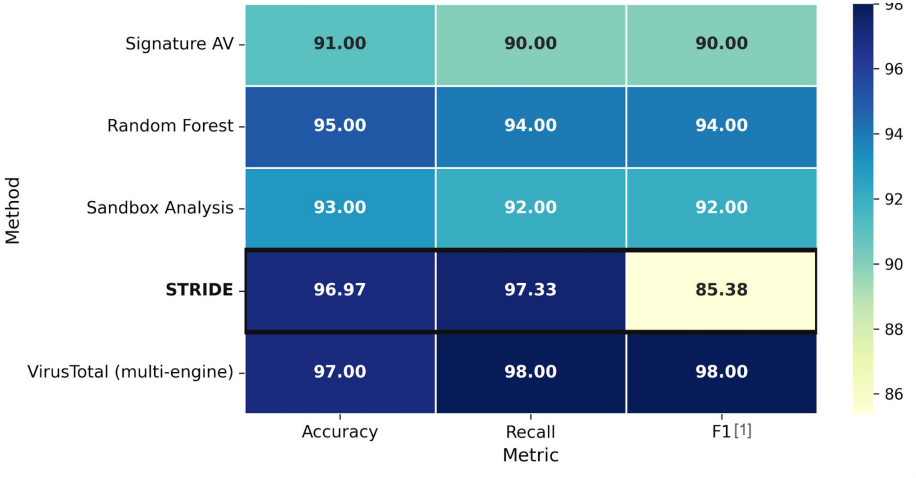

Figure 11: Malware Detection Method Comparison

Graphic created by student with matplotlib, 2026

Sources: AV-TEST Security Report (2024); Anderson et al. (2022); MITRE ATT&CK Evaluations (2023)

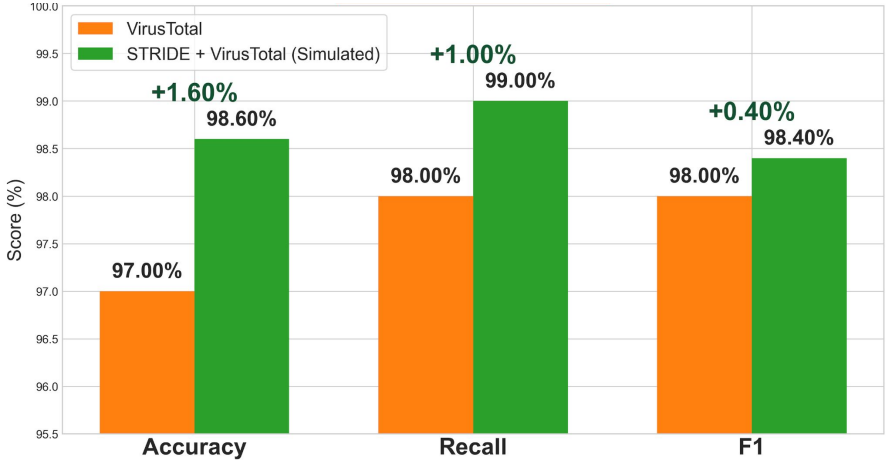

Figure 12: Performance of VirusTotal vs VirusTotal + STRIDE

Graphic created by student with matplotlib, 2026

[1] harmonic mean of precision and recall

Discussion

Interpretation of Results

- Stride achieved a competitive accuracy on test PDFs
- Lower F1 reflects stricter anomaly detection
- Structural features detect novel malware behavior

Challenge

- Difference in neighborhoods for artificial and web-scraped PDFs
  - Solution: Created two different benign manifolds

Result Justification

- Manifold Hypothesis: High-dimensional real-world data tend to lie on a low-dimensional manifold embedded in feature space. [1]

[1] Fefferman, C., Mitter, S., & Narayanan, H. (2016)

Conclusions & Next Steps

- Unsupervised detection **reduces reliance** on malware signatures
- Provides **meaningful** security signals
- Overall coverage of existing engines **improves** with STRIDE
- **Fresh method** for malware anomaly detection

Next Steps:

- Larger real-world dataset → reduces dataset bias
- **True zero-day evaluation → direct comparative results**
- Feature expansion → more omnescent

References/Acknowledgments

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Acknowledgements

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